

8D PROBLEM SOLVING REPORT

Quality Engineering & Technical Methodology — Automotive Engine Shaft Component

1. HEADER DETAILS & TARGET SPECIFICATIONS

OVERVIEW & SPECS

Report Title	8D Report: Automotive Engine Shaft — Oversized Diameter
Part Name & Number	Engine Shaft (Part No: ES-2026-X)
Target Specification	Actual: Diameter > 25.03 mm Standard / Ideal Target: 25.00 ± 0.02 mm <i>Max = 25.02 mm Min = 24.98 mm</i>

Specification Philosophy Comparison:

- **Taguchi Philosophy:** Specification must be as near to the target (25.00 mm) as possible to minimize quality loss.
- **Traditional Philosophy:** Any value within limits (24.98 mm < X < 25.02 mm) is accepted as pass.
- **Most Ideal Case:** 25.00 mm (Zero deviation from target center).

D1 — FORM CROSS-FUNCTIONAL TEAM (CFT)

TEAM STRUCTURE

1. Quality Inspection Department	Quality Inspector, QA/QC Lead Personnel
2. Production Department	Line Supervisors & Machining Technicians
3. Manufacturing Department	Process Engineers & Tooling Specialists
4. Management Department	Overall Cross-Functional Team Coordination & Handling
5. Operational Department	Shopfloor Logistics & Workflow Supervisors

D2 — DEFINE PROBLEM (5W2H FRAMEWORK)

PROBLEM DESCRIPTION

1. What	Diameter of automotive engine shaft exceeds specifications, measuring more than 25.03 mm (Standard: 25.00 ± 0.02 mm).
2. Where	Detected at Production Line / Quality Inspection Bench.
3. When	Detected when parts were about to be dispatched to customer.
4. Who	1st Level: Quality Inspector / QC Department 2nd Level: Customer feedback/escalation.
5. Why (Deviation)	Standard limit is 24.98 mm < X < 25.02 mm. Deviation observed: +0.01 mm above maximum allowable specification limit.
6. How Many	Rejection Rate increased from 2% → 8.5% (15% of manufactured parts are currently out of tolerance).

D3 — IMMEDIATE CONTAINMENT ACTION (ICA)

CONTAINMENT & ISOLATION

Standard Protocol (ISO 9001:2015 Clause 8.7 — Control of Non-Conforming Outputs):

- **Red Flag Tagging:** Immediately attach Red Flags / Tags to all defective and non-conforming parts to isolate them.
- **Line Halt:** Stop the production line immediately to prevent further manufacturing of out-of-tolerance parts.

D4 — IDENTIFY, VERIFY & VALIDATE ROOT CAUSE ANALYSIS (RCA)

ANALYSIS & DIAGNOSTICS

Step 1: Ishikawa Fishbone Diagram & 5 Why Analysis (6M Breakdown)

- **Man:** Unskilled operator or lack of training with respect to metrology, inspection procedures, and precision accuracy.
- **Material:** Excessively tough / hard material properties affecting tool cutting dynamics.
- **Measurement:** Metrological accuracy and precision incorrect → Uncalibrated measurement instruments used; defects present in precision measurement tools.
- **Machine:** Increased operational speed causing reduced precision & accuracy; excessive machine vibrations.
- **Environment:** Natural and ambient environmental errors affecting measurement thermal expansion.

Step 2: Applying Pareto Chart (80/20 Rule)

Prioritize root causes: **Machine / Man > Measurement > Material** (Addressing top 20% causes to eliminate 80% defect impact).

Step 3: Failure Mode & Effects Analysis (FMEA)

Conduct System FMEA + Design FMEA for machine across overall parameters (Man, Measurement, Machine, Material, Environment). Assign **Severity (S)**, **Occurrence (O)**, and **Detection (D)** numbers to calculate **RPN (Risk Priority Number)**.

Step 4: Six Sigma Methodology — Apply Six Sigma framework along with **DMAIC** (Define, Measure, Analyze, Improve, Control).

Step 5: Process Control & Prevention — Apply **PDCA Cycle** (Plan-Do-Check-Act) for preventing future failures, driving continuous improvement under the **Kaizen Philosophy**.

Step 6: Advanced Management Tools Integration — Apply Matrix Diagram, Process Decision Program Chart (PDPC), and Interrelationship Diagram. Combine PDCA + Kaizen while aligning with **Taguchi Philosophy** and **Philip Crosby's Zero Defect System**.

D5 — CHOOSE & VALIDATE PERMANENT CORRECTIVE ACTIONS (PCA)

ACTION
SELECTION

TQM Tools & Principles Application:

Integrate Six Sigma, DMAIC, PFMEA, PDCA Cycle, Control Charts (7 QC Tools: Pareto, Ishikawa, Histogram), and New Management Tools (Advanced Matrix, PDPC, Interrelationship Diagram).

Manufacturing & Design Solutions:

Utilize 3D Printing & Design, Rapid Prototyping, CAD Modeling, and Additive Manufacturing for system validation. For operational precision on the shopfloor, deploy upgraded **CNC Machining** controls.

D6 — IMPLEMENT & VALIDATE PERMANENT CORRECTIVE ACTIONS (PCA)

EXECUTION &
VALIDATION

- **Execution:** Implement chosen CNC machining modifications, recalibrate instruments, and train operating personnel.
- **Validation:** Use Statistical Quality Control tools:
 - Deploy **Control Charts** (*X-bar & R Charts*) for statistical process monitoring.
 - Evaluate Process Capability Indices C_p and C_{pk} to ensure process variation remains strictly within specification limits.

D7 — PREVENT RECURRENCE (SYSTEMIC PREVENTION)

SYSTEM STANDARDIZATION

1. **Reform Standard Operating Procedures (SOPs):** Update work instructions prioritizing core principles of Taguchi Philosophy and Zero Defect System.
2. **Overall System Auditing (ISO 19011 Guidelines):**
 - *Internal Audit:* Conducted periodically by internal company audit team.
 - *External Audit:* Conducted by third-party certification bodies / ISO agencies.
3. **Continuous Improvement:** Always embed Kaizen philosophy for ongoing incremental process refinements.
4. **System Benchmarking:** Implement External, Functional, and Cross-Functional Benchmarking across all operational lines.

- **Sign-off & Official Documentation:** Formally sign off 8D documentation with authorized signatures from QA/QC Lead and Quality Inspection (QI) Department.
- **Recognition & Motivation:** Empower team members for their contribution, dedication, and hard work using **Deming Award Methodology** for team motivation.
- **Leadership Philosophy:** Follow Deming's principle of providing authority, empowering team members, and striving to *"be a leader, not a boss."*